RATE ANALYSIS (2082/083)

Description of works	Providing and fixing of non reflective warning , mandatory and informatory sign board of 2mm thick MS sheet with back support frame on heavy 50mm tube or Channel section of 75mm *40mm firmly fixed to the ground by means of properly designed foundation with M 10/40 grade cement concrete 300mm *300mm *300mm as per drawings and Technical specifications /DoR Publication.	Unit = nos. (For 4 traffic signs nos.)
Spec. cl. No: 1501		

Norms no 15.1	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	2.00	855.00	1,710.00	excavation for foundation	cum	0.21	106.43	22.35	Tractor with trolly	hr	3.00	1,316.38	3,949.14
	Skilled	day	1.00	1,215.00	1,215.00	Cement concrete M10/40	cum	0.11	9,307.77	1,023.86					
						Paints two coats	sqm	3.51	478.74	1,680.38					
						Mild steel channel iron 75*40*6mm, 3.0 m long @ 6.8 kg/m incl. 5% wastage OR Heavy duty steel tube of intamal dia 50mm (6.19 kg/m), 3.0 m long including 5%	m	12.6	689.85	8,692.07					
						Angle iron 50*50*6mm for hold fast including 5% wastage	Kg	4.24	159.86	677.81					
						60 cm circular	sqm	1.2	2,770.62	3,324.74					
	Sub total A				2,925.00	Sub total of B =				15,421.20	Sub total of C =				3,949.14
	Sub total of A +B + C =				22,295.34	Contractor's overhead expenses 15% =				3,344.30	Total			25,639.64	
														Unit Rate	6,409.91

Description of works	Providing and fixing of non reflective warning , mandatory and informatory sign board of 2mm thick MS sheet with back support frame on heavy 50mm tube or Channel section of 75mm *40mm firmly fixed to the ground by means of properly designed foundation with M 10/40 grade cement concrete 300mm *300mm *300mm as per drawings and Technical specifications /DoR Publication.	Unit = nos. (For 4 traffic signs nos.)
Spec. cl. No: 1501		

Norms no 15.1	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	2.00	855.00	1,710.00	excavation for foundation	cum	0.21	106.43	22.35	Tractor with trolly	hr	3.00	1,316.38	3,949.14
	Skilled	day	1.00	1,215.00	1,215.00	Cement concrete M10/40	cum	0.11	9,307.77	1,023.86					
						Paints two coats	sqm	3.51	478.74	1,680.38					
						Mild steel channel iron 75*40*6mm, 3.0 m long @ 6.8 kg/m incl. 5% wastage OR Heavy duty steel tube of intamal dia 50mm (6.19 kg/m), 3.0 m long including 5%	m	12.6	689.85	8,692.07					
						Angle iron 50*50*6mm for hold fast including 5% wastage	Kg	4.24	159.86	677.81					
						80cm*60 cm rectangular	sqm	2.02	2,770.62	5,596.65					
	Sub total A				2,925.00	Sub total of B =				17,693.11	Sub total of C =				3,949.14
	Sub total of A +B + C =				24,567.25	Contractor's overhead expenses 15% =				3,685.09	Total			28,252.33	
														Unit Rate	7,063.08

RATE ANALYSIS (2082/083)

Description of works	Providing and installation of 150 mm * 150 mm 1. 5 m long delineators (road way indicators, hazard markers, object markers), 80-100 cm high above ground level, painted black and white in 20 cm wide strips, buried or pressed into the ground and conforming to the drawings and Technical Specifications.	Unit = nos. (For 30 nos.)
Spec. cl. No: 1507		

Norms no 15.12	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	7.00	855.00	5,985.00	excavation for foundation	cum	0.47	106.43	50.02	Tractor with trolly	hr	3.00	1,316.38	3,949.14
	Skilled	day	1.00	1,215.00	1,215.00	Cement concrete M15/40	cum	1.01	11,169.99	11,281.69					
						Paints two coats	sqm	14.4	478.74	6,893.86					
						Steel reinforcement as per standard drawing (4 nos 8 mm dia and 11	Kg	112.8	91.00	10,264.80					
	Sub total A				7,200.00	Sub total of B =				28,490.37	Sub total of C =				3,949.14
	Sub total of A +B + C =				39,639.51	Contractor's overhead expenses 15% =				5,945.93	Total			45,585.44	
														Unit Rate	1,519.51

Description of works	Providing and Fixing Reinforced cement concrete M15 grade kilometer Post including painting and printing as per Standard Drawing-2070 and Technical Specifications. Position (I) Five Kilometer Post Precast	Unit = nos. (For 6 nos.)
Spec. cl. No: 1506		

Norms no 15.11 (i)	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	6.00	855.00	5,130.00	excavation for foundation	cum	1.2	106.43	127.72	Tractor with trolly	hr	3.00	1,316.38	3,949.14
	Skilled	day	1.00	1,215.00	1,215.00	Cement concrete M15/40	cum	1.2	11,169.99	13,403.99					
						Cement concrete M10/40	cum	1.2	9,307.77	11,169.33					
						Paints two coats	sqm	10.2	478.74	4,883.15					
						Lettering on Km Post	cm-letter	1800	5.00	9,000.00	LS rate				
						Steel reinforcement as per standard	Kg	63.6	91.00	5,787.60					
	Sub total A				6,345.00	Sub total of B =				44,371.78	Sub total of C =				3,949.14
	Sub total of A +B + C =				54,665.92	Contractor's overhead expenses 15% =				8,199.89	Total			62,865.81	
														Unit Rate	10,477.64

Description of works	Providing and Fixing Reinforced cement concrete M15 grade kilometer Post including painting and printing as per Standard Drawing-2070 and Technical Specifications. Position (I) One Kilometer Post Precast	Unit = nos. (For 14 nos.)
Spec. cl. No: 1506		

Norms no 15.11 (ii)	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	7.00	855.00	5,985.00	excavation for foundation	cum	2.38	106.43	253.31	Tractor with trolly	hr	3.00	1,316.38	3,949.14
	Skilled	day	2.00	1,215.00	2,430.00	Cement concrete M15/40	cum	1.4	11,169.99	15,637.99					
						Cement concrete M10/40	cum	2.38	9,307.77	22,152.50					
						Paints two coats	sqm	11.9	478.74	5,697.01					
						Lettering on Km Post	cm-letter	1680	5.00	8,400.00	LS rate				
						Steel reinforcement as per standard	Kg	82.88	91.00	7,542.08					
	Sub total A				8,415.00	Sub total of B =				59,682.88	Sub total of C =				3,949.14
	Sub total of A +B + C =				72,047.02	Contractor's overhead expenses 15% =				10,807.05	Total			82,854.08	
														Unit Rate	5,918.15

RATE ANALYSIS (2082/083)

Description of works	Providing and laying of Road Marking with Hot Applied Thermoplastic Compound at least 2mm thick including Reflectorizing Glass Beads on Bituminous Surface As per DOR Traffice sign manual/Specification.The finished surface to be level, uniform and free from streaks and holes. (i) (on smooth surface (similar to Asphalt concrete and rigid pavements))	Unit = sqm (For 400 sqm.)
Spec. cl. No: 1504		

Norms no 15.9 (i)	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	4.00	855.00	3,420.00	Hot applied thermoplastic compound	liter	930	230.00	213,900.00	Tractor with trolly	hr	10.00	1,316.38	13,163.80
	Skilled	day	2.00	1,215.00	2,430.00	reflective glass beads	Kg	100	50.00	5,000.00	Road marking machine	hr	# # #	835.00	8,350.00
	Sub total A				5,850.00	Sub total of B =				218,900.00	Sub total of C =				21,513.80
	Sub total of A +B + C =				246,263.80	Contractor's overhead expenses 15% =				36,939.57	Total			283,203.37	
														Unit Rate	708.01

Description of works	Providing and laying of Road Marking with Hot Applied Thermoplastic Compound at least 2mm thick including Reflectorizing Glass Beads on Bituminous Surface As per DOR Traffice sign manual/Specification.The finished surface to be level, uniform and free from streaks and holes. (i) (on rough surface (similar to surface dressing))	Unit = sqm (For 300 sqm.)
Spec. cl. No: 1504		

Norms no 15.9 (ii)	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	4.00	855.00	3,420.00	Hot applied thermoplastic compound	liter	1200	230.00	276,000.00	Tractor with trolly	hr	10.00	1,316.38	13,163.80
	Skilled	day	2.00	1,215.00	2,430.00	reflective glass beads	Kg	150	50.00	7,500.00	Road marking machine	hr	# # #	835.00	8,350.00
	Sub total A				5,850.00	Sub total of B =				283,500.00	Sub total of C =				21,513.80
	Sub total of A +B + C =				310,863.80	Contractor's overhead expenses 15% =				46,629.57	Total			357,493.37	
														Unit Rate	1,191.64

Description of works	Provide and erecting "W" metal beam crash barrier comprising of 3.00mm thick corrugated sheet metal beam rail, 70cm above road/ground level,fixed on ISMC series channel vertical post, 150*75*5 mm spaced 2m c/c,1.8m high,1.1m below ground/road level,,metal beam rail to be fixed on the vertical post with a spacer of chaneel section 150*75*5mm 330mm long complete as per Drawing and Technical specifications	Unit = meter (For 40 m length.)
Spec. cl. No: 1509		

Norms no 15.14 (A)	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	10.00	855.00	8,550.00	Hot dip galvanized corrugated W beam sheet 3 mm thick	kg	563.61	139.79	78,784.22	Tractor with trolly	hr	3.00	1,316.38	3,949.14
	Skilled	day	1.00	1,215.00	1,215.00	Hot dip galvanized channel post 150*75*5	kg	695.52	279.57	194,446.53					
						Hot dip galvanized spacer channel 150*75*5	kg	127.51	279.57	35,647.97					
						M20 concrete	cum	0.99	12,292.17	12,169.25					
						EW excavation	cum	0.99	106.43	105.37					
						add 25 % of the cost of material for fabrication , nuts bolts and washers etc)				-					
	Sub total A				9,765.00	Sub total of B =				321,153.34	Sub total of C =				3,949.14
	Sub total of A +B + C =				334,867.48	Contractor's overhead expenses 15% =				50,230.12	Total			385,097.60	
														Unit Rate	9,627.44

RATE ANALYSIS (2082/083)

Description of works	Construction of Subgrade and Earthen Shoulders with approved Material (capping layer) Providing and laying sub-grade and earthen shoulders with approved Material obtained from borrow pits with all lifts & leads as per Drawing and Technical Specifications.	Unit = cum (For 600 cum)
Spec. cl. No: 1004		

Norms no 10.3	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	6.00	855.00	5,130.00	Cost of water	liter	72000	0.19	13,860.00	Motor	hr	6.00	3,957.50	23,745.00
	Skilled	day	1.00	1,215.00	1,215.00	capping layer material	cum	750	1,184.00	888,000.00	Vibratory roller	hr	6.00	2,626.11	15,756.66
	Sub total A				6,345.00	Sub total of B =				901,860.00	Sub total of C =				39,501.66
	Sub total of A +B + C =				947,706.66	Contractor's overhead expenses 15% =				142,156.00	Total			1,089,862.66	
														Unit Rate	1,816.44

Description of works	Providing and mixing of Anti stripping agent as per Design/ direction of Engineer	Unit = Kg (For 200 kg)
Spec. cl. No:1300		

Norms no 13.16	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
	Unskilled	day	1.00	855.00	855.00	Additive material	kg	210	546.48	114,760.80	Add 3 percent of Labour		1.00	26.01	26.01
	Skilled	day	0.01	1,215.00	12.15					-					-
	Sub total A				867.15	Sub total of B =				114,760.80	Sub total of C =				26.01
	Sub total of A +B + C =				115,653.96	Contractor's overhead expenses 15% =				17,348.09	Total			133,002.06	
														Unit Rate	665.01

EXTRA ITEM

Description of works:	Providing and filling stone/ boulder in gabion boxes/ mattresses etc including dressing, bedding, bonding all complete as per drawing and technical specification	Unit : 10 m3
Spec. cl. No: 2401		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
24.4	Skilled	md	2.00	1215.00	2430.00	Boulder/Stone	m3	11.00	2218.00	24398.00					
	Unskilled	md	8.00	855.00	6840.00										
				Sub total of A =	9270.00				Sub total of B =	24398.00				Sub total of C =	0.00
				Sub total of A +B + C =	33668.00	Contractor's overhead expenses 15% =				5,050.20				Unit Rate =	38718.20
														Rate per m3	3871.82

Description of works:	Maintenance of Earth Shoulder (stripping excess soil) Strippingexcesssoil from theshoulder surfaceto achieve the approved level and compacting as per Technical Specifications and direction of the Engineer.	Unit : 100 m2
Spec. cl. No: 2906		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
29.80	Skilled	md	0.10	1215.00	121.50						Plate compactor	hr	6	214	1284.00
	Unskilled	md	4.00	855.00	3420.00										
				Sub total of A =	3541.50				Sub total of B =	0.00				Sub total of C =	1284.00
				Sub total of A +B + C =	4825.50	Contractor's overhead expenses 15% =				723.83				Unit Rate =	5549.33
														Rate per m2	55.49

Description of works:	Land Slide Clearance in soil Clearance of land slides in soil and ordinary rock by machine and disposal of the same on the valley side	Unit :500 m3
Spec. cl. No: 2900,900		

Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
29.19	Skilled	md	1.00	1215.00	1215.00						Dozer/loader/Excavator	hr	6	3265.15	19590.90
	Unskilled	md	3.00	855.00	2565.00										
				Sub total of A =	3780.00				Sub total of B =	0.00				Sub total of C =	19590.90
				Sub total of A +B + C =	23370.90	Contractor's overhead expenses 15% =				3,505.64				Unit Rate =	23076.54

RATE ANALYSIS (2082/083)

														Rate per m3	53.75
Description of works:	Hill Side Drain Clearance Removal of earth from the choked hill side drain and disposing it on the valley side manually														Unit :10 m
Spec. cl. No: 2900,900															
Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
29.18	Skilled	md	1.00	1215.00	1215.00										0.00
	Unskilled	md	1.00	855.00	855.00										
				Sub total of A =	2070.00				Sub total of B =	0.00				Sub total of C =	0.00
				Sub total of A +B + C =	2070.00			Contractor's overhead expenses 15% =		310.50				Unit Rate =	2380.50
														Rate per m	238.05
Description of works:	Providing and laying of Machine mixed Plain/Reinforced Cement Concrete in Foundation complete as per Drawing and Technical Specifications. Grade M15 (Excluding Formwork)														Unit : 15 m3
Spec. cl. No: 2000															
Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
20.2 A	Skilled	md	3.00	1215.00	3645.00	cement	mt	4.130	13500.00	55755.00	mixer (0.28/ 0.20 m^3)	hr	6.00	826.26	4957.56
	Unskilled	md	30.00	855.00	25650.00	aggregate 40mm	m^3	8.100	1990.00	16119.00	Generator	hr	6.00	1173.44	7040.64
						aggregate 20mm	m^3	4.050	1990.00	8059.50	Mix Design and Quality Control		0.00%	of Cost of concrete	0.00
						aggregate 10mm	m^3	1.350	2044.00	2759.40					
						Coarse Sand	m^4	6.750	2329.00	15720.75					
						water	lit	2000.000	0.19	385.00					
				Sub total of A =	29295.00				Sub total of B =	98798.65				Sub total of C =	11998.20
				Sub total of A +B + C =	140091.85			Contractor's overhead expenses 15% =		21,013.78				Total Rate =	161,105.63
														Rate per m3	10740.38
Description of works:	Providing and laying of Plain/Reinforced Cement Concrete in Foundation complete as per Drawing and Technical Specifications. Grade M15 (Excluding Formwork) Manual														Unit : 15 m3
Spec. cl. No: 2000															
Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
20.2 A	Skilled	md	4.50	1215.00	5467.50	cement	mt	4.130	13500.00	55755.00	mixer (0.28/ 0.20 m^3)	hr	6.00	826.26	4957.56
	Unskilled	md	45.00	855.00	38475.00	aggregate 40mm	m^3	8.100	1990.00	16119.00	Generator	hr	6.00	1173.44	7040.64
						aggregate 20mm	m^3	4.050	1990.00	8059.50	Mix Design and Quality Control		0.00%	of Cost of concrete	0.00
						aggregate 10mm	m^3	1.350	2044.00	2759.40					
						Coarse Sand	m^4	6.750	2329.00	15720.75					
						water	lit	2000.000	0.19	385.00					
				Sub total of A =	43942.50				Sub total of B =	98798.65				Sub total of C =	11998.20
				Sub total of A +B + C =	154739.35			Contractor's overhead expenses 15% =		23,210.90				Total Rate =	177950.2525
														Rate per m3	11863.35
Description of works:	Providing and laying of Plain/Reinforced Cement Concrete in Foundation complete as per Drawing and Technical Specifications. PCC Grade M20 (Excluding Formwork)														Unit : 15 m3
Spec. cl. No: 2000															
Norms No.	Labour (A)					Material (B)					Equipment (C)				
	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount	Type	Unit	Qty.	Rate	Amount
20.2 B	Skilled	md	3.00	1215.00	3645.00	cement	mt	5.160	13500.00	69660.00	mixer (0.28/ 0.20 m^3)	hr	6.00	826.26	4957.56
	Unskilled	md	30.00	855.00	25650.00	aggregate 40mm	m^3	5.400	1990.00	10746.00	Generator	hr	6.00	1173.44	7040.64
						aggregate 20mm	m^3	5.400	1990.00	10746.00					